A

Project Report

On

Online Car MarketPlace

More Harshad Mohan

Roll No.: 24240

MCA–I

SEM–II

Under the guidance of

Prof. Ashwini Mohite
For the Academic Year 2024-25

Sinhgad Institutes

Sinhgad Technical Education Society's

Sinhgad Institute of Management

Vadgaon Bk Pune 411041

(Affiliated to SPPU Pune & Approved by AICTE New Delhi)

SINHGAD TECHNICAL EDUCATION SOCIETY'S
SINHGAD INSTITUTE OF MANAGEMENT®
Sinhgad Institutes
Affiliated to Savitribai Phule Pune University, Approved by AICTE, New Delhi

Prof. M. N. Navale
M.E. (ELECT.), MIE, MBA
FOUNDER PRESIDENT

Dr. (Mrs.) Sunanda M. Navale
B.A., MPM, Ph.D
FOUNDER SECRETARY

Dr. Chandrani Singh
MCA, ME, (Com. Sci.), Ph.D
DIRECTOR - MCA

CERTIFICATE

Date:

This is to certify that Mr More Harshad Mohan has successfully completed his project work entitled "Online Car MarketPlace" in partial fulfillment of MCA – I SEM –II Mini Project for the year 2024-2025. He has worked under our guidance and direction.

Prof Ashwini Mohite
Project Guide

Dr. Chandrani Singh
Director, SIOM-MCA

Examiner 1

Examiner 2

Date:

Place: Pune

Celebrating 25 Years
OF ACADEMIC EXCELLENCE

S. No. 44/1, Vadgaon (Bk.), Off Sinhgad Road, Pune - 411 041 | **Tel.:** (020) 24358360 / 24356592 **Fax:** (020) 24355531
E-mail : directormca_siom@sinhgad.edu, **Website :** www.sinhgad.edu

DECLARATION

I certify that the work contained in this report is original and has been done by me under the guidance of my guide.

- The work has not been submitted to any other Institute for any degree or diploma.
- I have followed the guidelines provided by the Institute in preparing the report.
- I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the report and giving their details in the references.

Name and Signature of Project Team Members:

Sr. No.	Seat No.	Name of students	Signature of students
1	24240	More Harshad Mohan	

ACKNOWLEDGEMENT

It is very difficult task to acknowledge all those who have been of tremendous help in this project. I would like to thank my respected guide Prof Ashwini Mohite for providing me necessary facilities to complete my project and also for their guidance and encouragement in completing my project successfully without which it wouldn't be possible. I wish to convey my special thanks and immeasurable feelings of gratitude towards Dr. Chandrani Singh, Director, SIOM-MCA. I wish to convey my special thanks to all teaching and non-teaching staff members of Sinhgad Institute of Management, Pune for their support.

Thank You

Yours Sincerely,

More Harshad Mohan

INDEX

Sr. No.	Chapter	Page No.
1	CHAPTER 1: INTRODUCTION	
1.1	Abstract	
1.2	Existing System and Need for System	
1.3	Scope of System	
1.4	Operating Environment Hardware and Software	
1.5	Brief Description of Technology Used	
2	CHAPTER 2: PROPOSED SYSTEM	
2.1	Feasibility Study	
2.2	Objectives of the proposed system	
2.3	Users of the system	
3	CHAPTER 3: ANALYSIS AND DESIGN	
3.1	Entity Relationship Diagram (ERD)	
3.2	Class Diagram	
3.3	Use Case Diagrams	
3.4	Activity Diagram	
3.5	Sequence Diagram	
3.6	Component Diagram	
3.7	Module and Hierarchy Diagram	
3.8	Table Design	
3.9	Data Dictionary	
3.10	Sample Input and Output Screens (Screens must have valid data. All reports must have at-least 5 valid records.)	
4	CHAPTER 4: CODING Sample code	
5	CHAPTER 5: LIMITATIONS OF SYSTEM	
6	CHAPTER 6: PROPOSED ENHANCEMENTS	
7	CHAPTER 7: CONCLUSION	
8	CHAPTER 8: BIBLIOGRAPHY	

INTRODUCTION

1.1 Abstract

The Integrated Hostel and Mess Solution is a comprehensive web-based application designed to manage hostel and mess operations efficiently. This system caters to the needs of students, hostel administrators, mess owners, and other stakeholders by providing a centralized platform for managing hostel rooms, meal menus, attendance, bookings, payments, and more. The application streamlines the daily operations of hostels and mess services, ensuring smooth coordination between the hostel management and the students while offering real-time updates and a seamless experience. By utilizing modern technologies and providing easy access to users, this system aims to simplify hostel and mess management for academic institutions.

1.2 Existing System and Need for System

Currently, hostel and mess management in educational institutions often relies on manual processes or separate systems for each service, leading to inefficiencies and errors. These systems usually lack integration, making it difficult for hostel managers, students, and mess owners to coordinate effectively. There are challenges like room allocation issues, mismatched meal bookings, inconsistent attendance tracking, and delays in payment processing.

1.3 Scope of System

- User Management: Different types of users, such as students, mess owners, hostel administrators, and staff, can register, log in, and manage their information.
- Room Allocation: Admins can allocate rooms to students based on their preferences and availability.
- Mess Menu Management: Mess owners can update meal menus for different days, and students can book their meals accordingly.
- Meal Booking System: Students can book meals for breakfast, lunch, and dinner based on available menus and track meal consumption.
- Attendance Tracking: Hostels can track the daily attendance of students.

1.4 Operating Environment Hardware and Software:

Server-side requirement

Software Requirement	Hardware Requirement
Operating System: Windows 7 or above	Processor: Intel Core i3 or above
Front End: Html,css,js	RAM: 4GB or above
Back End: Python , Django	HDD: 512 GB or above

Database: MySql	
Web Browser: Chrome ,Mozilla ,Firefox	

Client-side requirement

Software Requirement	Hardware Requirement
Operating System: Windows 7 or above	Processor: Intel Core i3 or above
Web Browser: Chrome, Mozilla Firefox	RAM: 2GB or above
	HDD: 512 GB or above

1.5 Brief Description of Technology Used

- Django Framework: The system is built using the Django web framework, which is a high-level Python framework that allows for the rapid development of secure and maintainable websites. Django's built-in features, such as the ORM (Object-Relational Mapping), authentication system, and admin interface, make it an ideal choice for web-based applications like the Integrated Hostel and Mess Solution.
- Python: The programming language used to build the backend logic of the system. Python is widely recognized for its simplicity, readability, and scalability, making it perfect for rapid web development.
- HTML5, CSS3, and JavaScript: These technologies are used to design the frontend of the application. HTML provides the structure, CSS is used for styling, and JavaScript (along with jQuery) handles user interaction and dynamic content updates.
- SQLite/PostgreSQL: The database management system used to store all the data related to user accounts, room allocations, meal bookings, attendance, and payments. SQLite is used for local development, while PostgreSQL is preferred for deployment in production environments due to its scalability and robustness.

PROPOSED SYSTEM

2.1 Feasibility Study

Technical Feasibility

The system will be built using Django, a robust Python-based web framework, which has all the necessary features for developing a web application. The backend logic is easy to develop, maintain, and scale, making it technically feasible. The system is

designed to run on a variety of devices, including desktop computers, laptops, and mobile devices. The backend infrastructure will require a server with sufficient processing power to handle multiple users and requests simultaneously.

Economic Feasibility

The development cost of the project will mainly involve the cost of the server and domain hosting, software licenses (if any), and the salaries of developers and designers. Given the use of open-source technologies like Django, Python, and PostgreSQL, the financial cost of software development will be relatively low.

Operational Feasibility

The Integrated Hostel and Mess Solution will streamline the operations of both the hostel and mess management, reducing manual efforts and human errors. The automation of room allocation, meal bookings, attendance tracking, and payment processing will enhance efficiency and coordination between hostel staff, mess owners, and students.

User interfaces will be simple and intuitive, making it easy for all users (students, hostel administrators, mess owners, etc.) to interact with the system.

Legal Feasibility:

The system complies with data protection and privacy regulations (such as GDPR or local laws regarding student data). The data will be stored securely, and sensitive information (such as payment details) will be encrypted.

Usage terms and privacy policies will be in place to protect both the users and the service providers (hostels and messes).

2.2 Objectives of the Proposed System

Streamline Hostel and Mess Operations:

- The primary objective is to create an integrated platform that combines the management of hostel rooms, mess services, meal bookings, attendance, and payments into one cohesive system.

Ease of Access for Users:

- The system will provide a user-friendly interface for students, hostel administrators, and mess owners. Students will be able to manage room allocations, meal bookings, and payments through a web-based platform.

Reduce Manual Efforts and Errors:

- By automating room assignments, meal menu updates, attendance tracking, and payment processing, the system aims to reduce human errors and administrative workload. This will increase operational efficiency and improve accuracy.

2.3 Users of the System

Students:

- Role: The primary users of the system who will interact with the portal to manage their hostel and mess-related activities.
- Responsibilities:
  - Register and update their personal details. o Book and manage hostel rooms. o View and book meals from the mess.
  - View and pay bills for hostel accommodation and mess services.
  - Check attendance and room availability.

Admin:

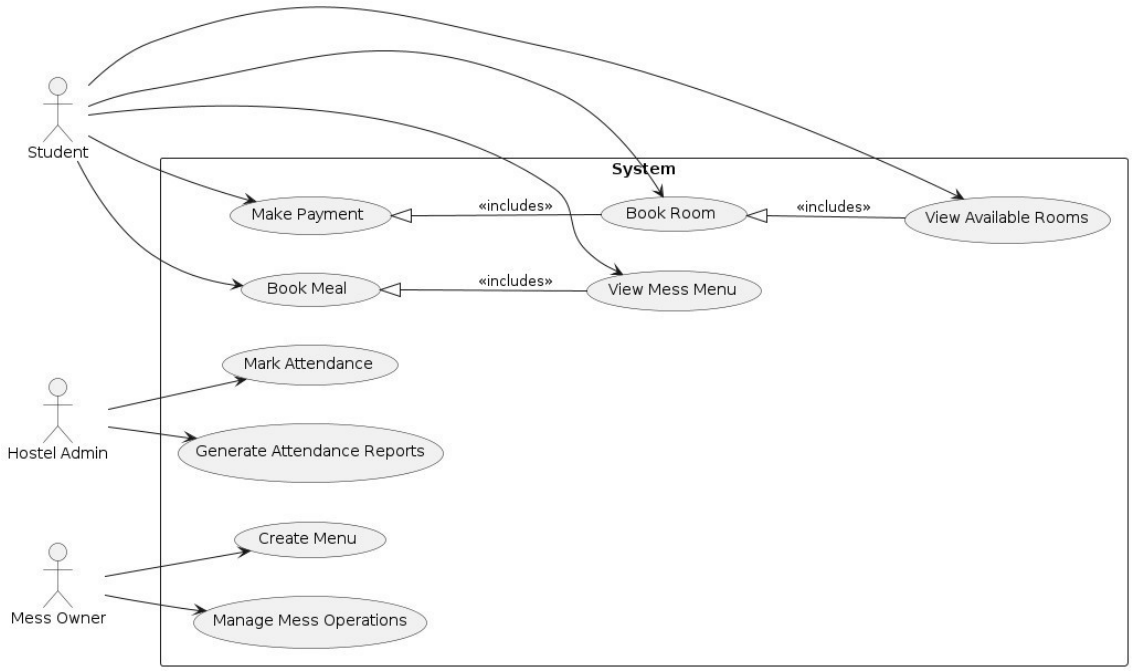
- Role: The superuser who manages the overall system.
- Responsibilities:
  - Maintain system configurations and manage user accounts. o Monitor student attendance, meal bookings, and payments.
  - Generate reports and analytics related to hostel and mess operations. o Perform backups and system maintenance.

Mess Owners/Managers:

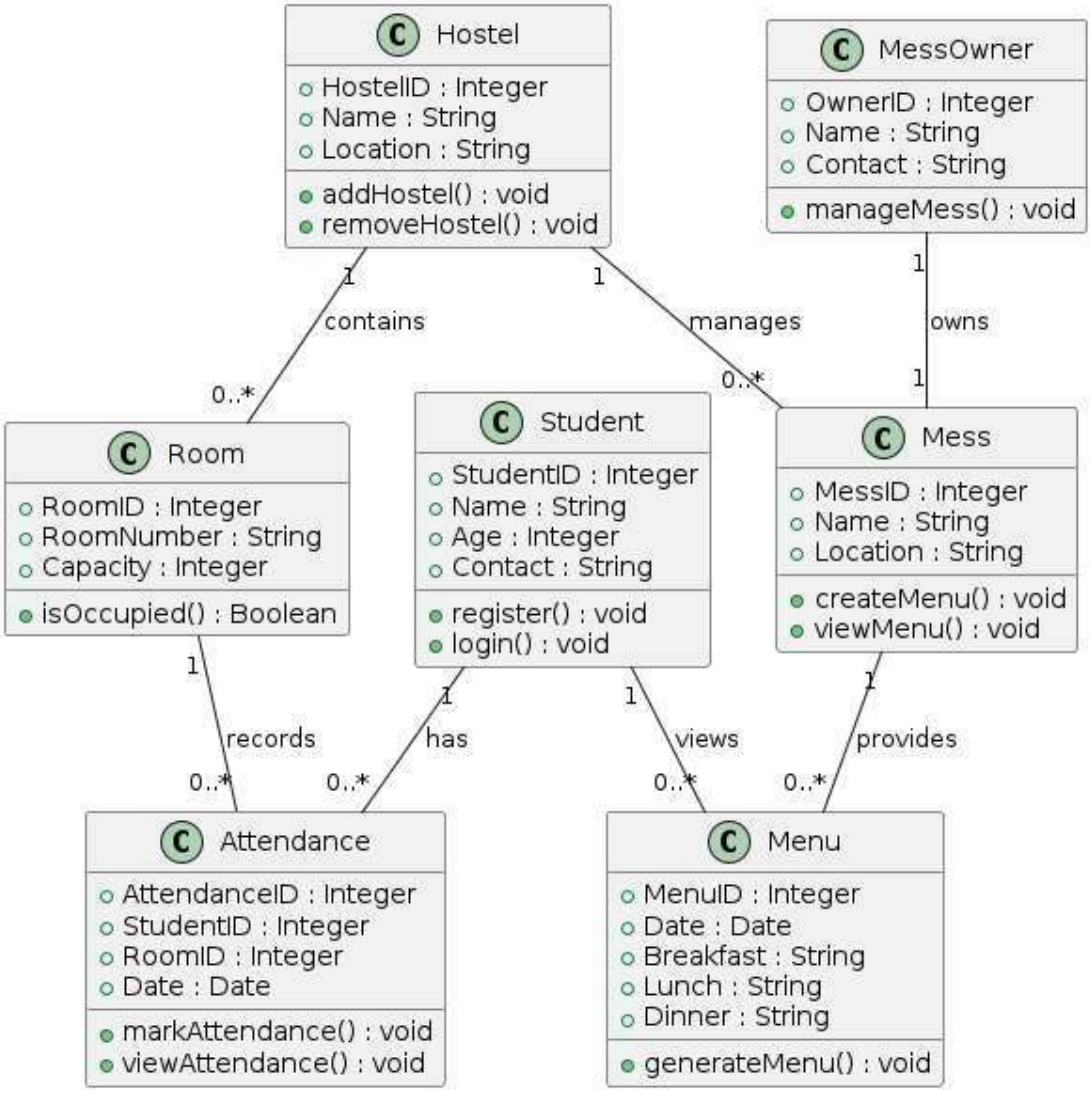
- Role: The users responsible for managing the mess services, meal menus, and meal bookings.
- Responsibilities:
  - Define and update meal menus for breakfast, lunch, and dinner.
  - Approve meal bookings made by students.
  - Track meal consumption and handle payments for mess services.

ANALYSIS AND DESIGN

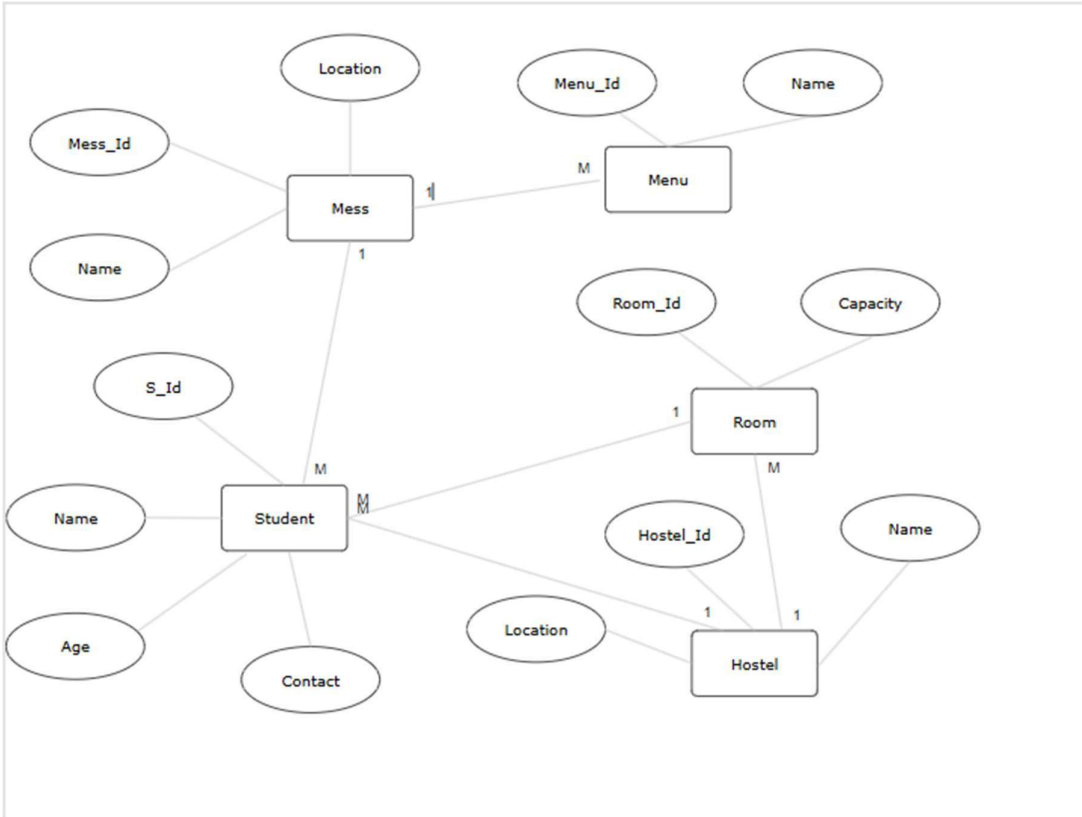
3.1 Use Case Diagram



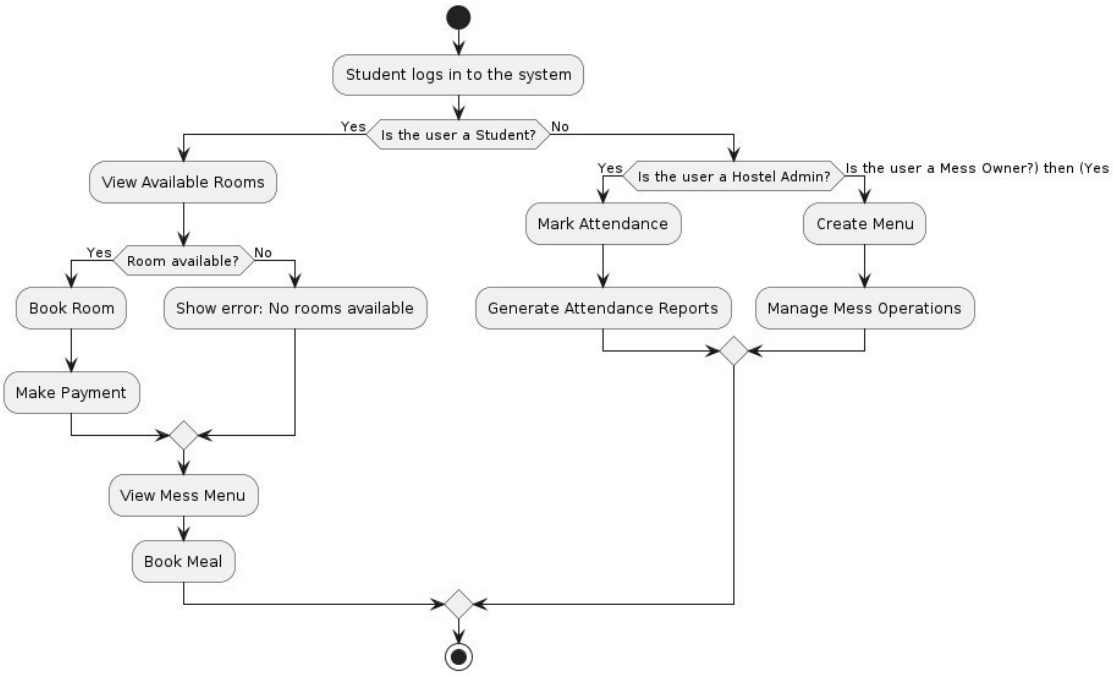
3.2 Class Diagram



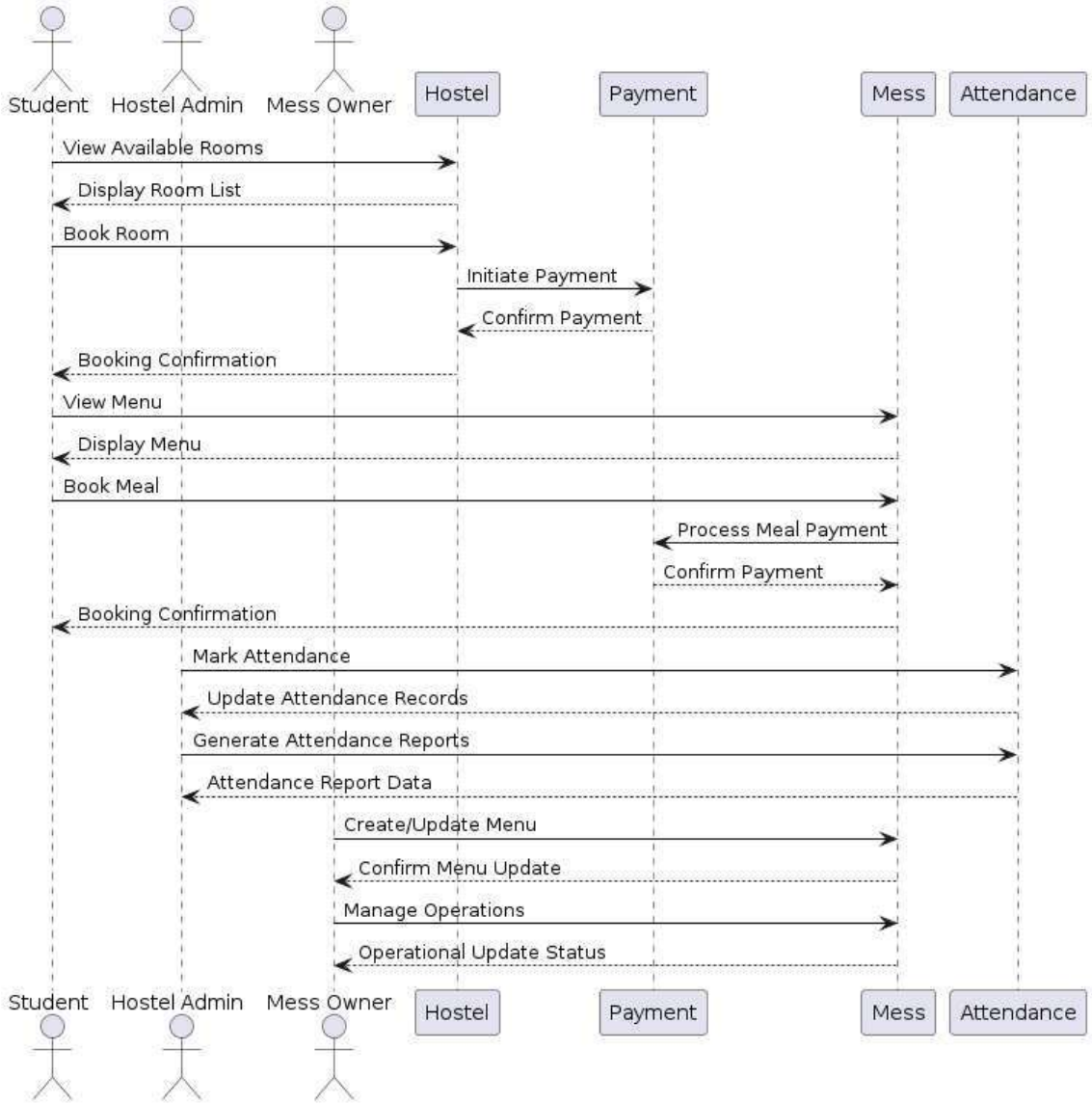
3.3 ER-Diagram



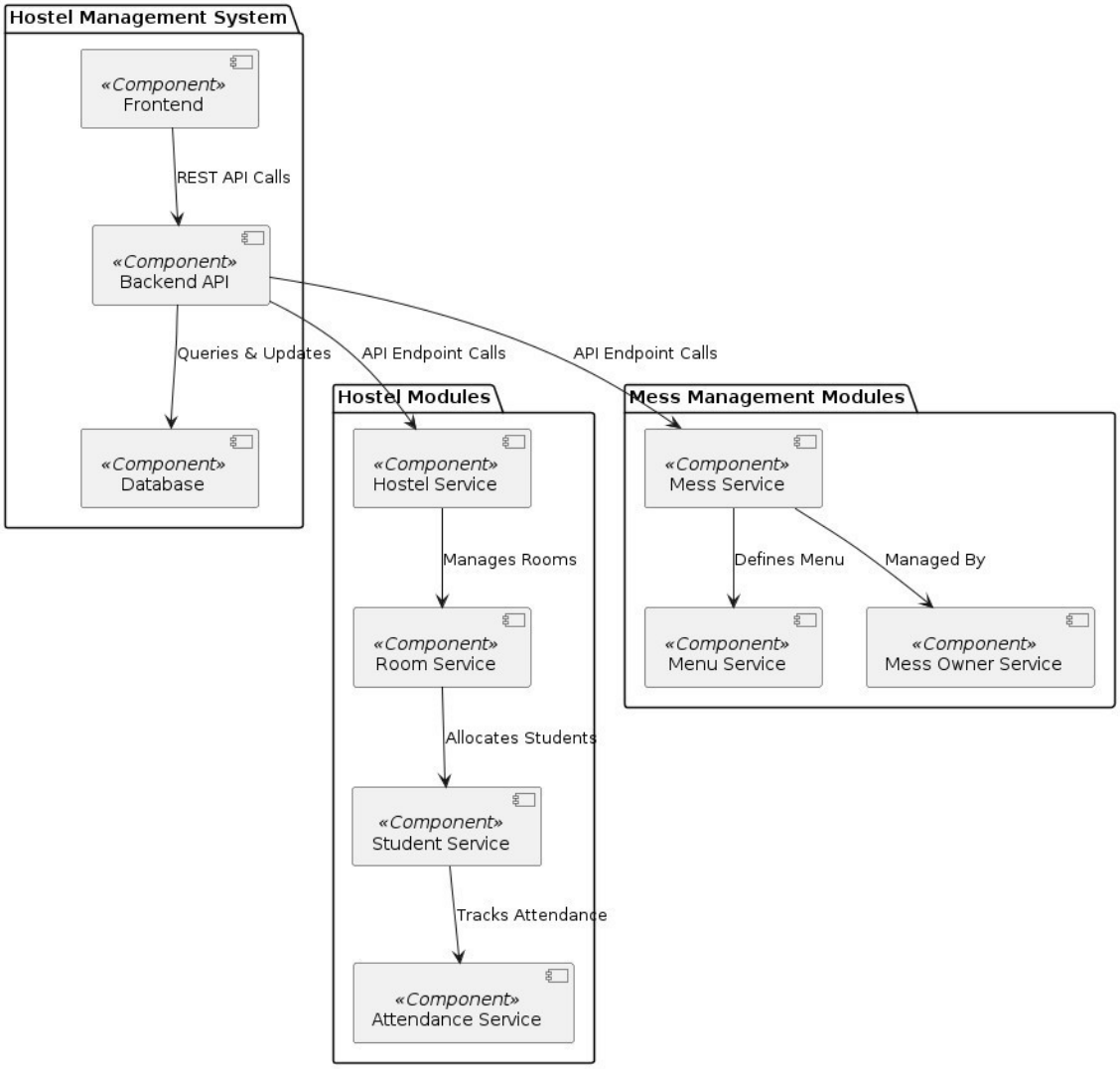
3.4 Activity Diagram



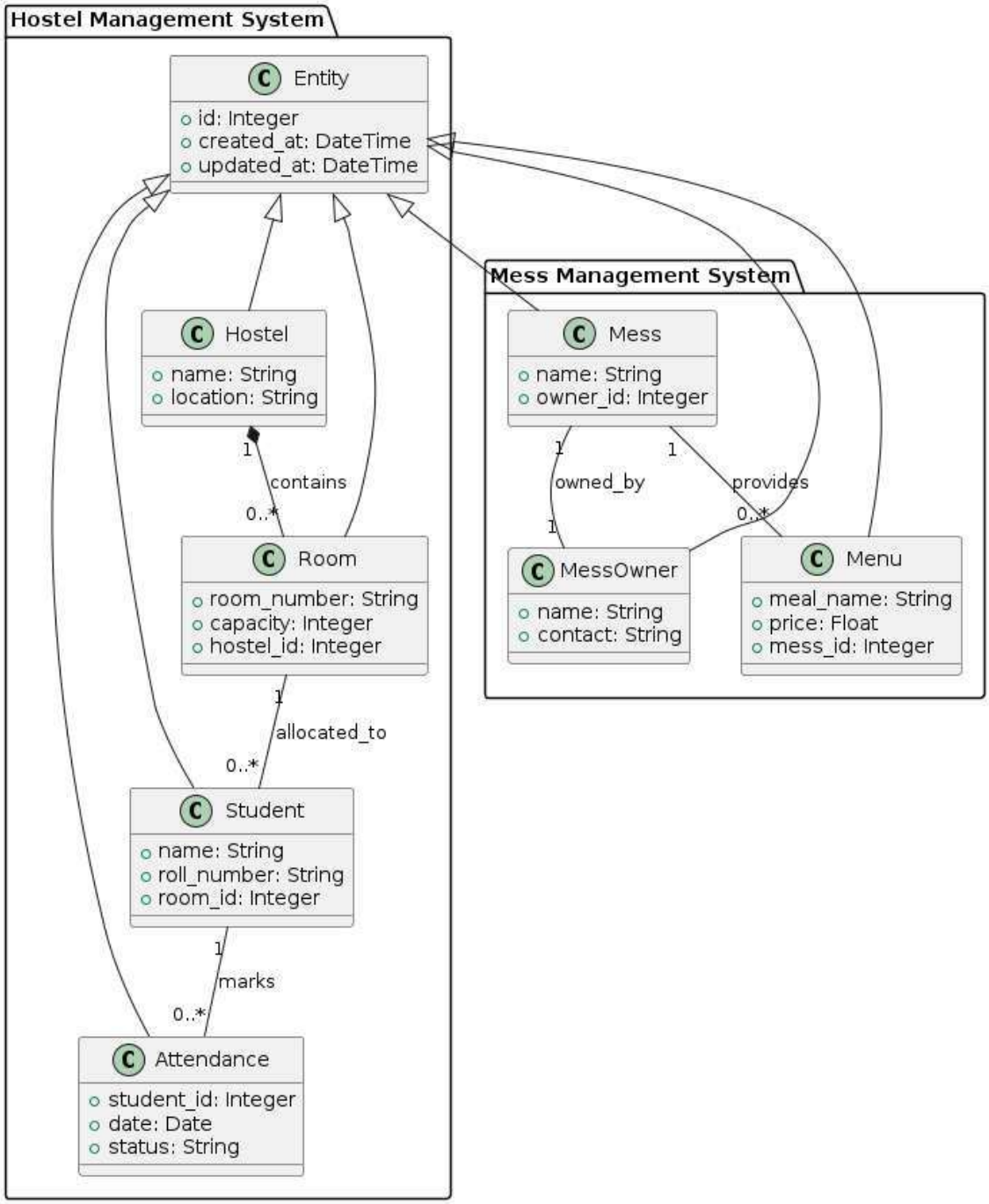
3.5 Sequence Diagram



3.6 Component Diagram



3.7 Heriarchy Diagram



3.8 Sample Input and Output Screens

File C:/HMP/d.html#hostel-info

Home Hostel Info Mess Info Contact

HOSTEL AND MESS PORTAL

Welcome to the Hostel and Mess Portal. Here you can access important information regarding hostel allocations and mess services. Use the buttons below to navigate to the respective sections.

Hostel Information Mess Information

Hostel Admission Information

Our hostel provides a safe, comfortable, and student-friendly environment, ensuring a balance between academics and personal growth. Below are the key features and details:

- Total Rooms: 100
- Fees Per Student: Rs. 20,000 per semester
- Available Rooms: 30
- Features:
  - 24/7 Security and Surveillance
  - Fully Furnished Rooms with Wi-Fi
  - On-site Laundry Facilities
  - Recreational Areas and Sports Facilities
  - Hygienic and Nutritious Food

Mess Information

Our mess provides hygienic and nutritious meals designed to suit the tastes and preferences of students. Below are the features of our mess services:

- Healthy and Balanced Meal Plans
- Breakfast, Lunch, and Dinner Provided Daily
- Special Meals on Weekends and Festivals
- Dedicated Feedback System for Meal Improvements
- Separate Vegetarian and Non-Vegetarian Meal Options

Enjoy delicious and homely meals that keep you energized throughout the day.

Apply Now Check Menu

CODING

```
hostel_management/
│
├── core/
│   ├── __init__.py
│   ├── admin.py
│   ├── apps.py
│   ├── migrations/
│   ├── models.py
│   ├── tests.py
│   ├── views.py
│   ├── forms.py
│   └── urls.py
│
├── hostel_management/
│   ├── __init__.py
│   ├── settings.py
│   ├── urls.py
│   └── wsgi.py │
├── manage.py
└── templates/
    ├── base.html
    ├── register.html
    ├── login.html
    ├── room_allocation.html
    ├── meal_booking.html
    ├── payment_view.html
    ├── attendance_report.html
    └── mark_attendance.html
```

Dashboard.html

```
<!DOCTYPE html>
<html lang="en">
```

```
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initialscale=1.0">
   <title>Hostel and Mess Portal Dashboard</title>
   <link rel="stylesheet" href="styles.css">
   <style>
body {
        font-family:        'Roboto',        sans-serif;
background-color: #f4f7f6; margin: 0;
        padding: 0;
      }
      .container {           max-
width: 1200px;
margin: 20px auto;
padding: 20px;
        background: linear-gradient(to right, #ffffff, #eef2f3);
box-shadow: 0 8px 16px rgba(0, 0, 0, 0.2);           border-radius:
12px;
      }
      header {           text-
align: center;           margin-
bottom: 20px;
      }
      header h1 {           font-size:
2.5em;           color: #007bff;
margin-bottom: 10px;           text-
transform: uppercase;
      }
      header p {
font-size: 1.2em;
color: #495057;
      }
nav {
```

```
            background-color: #333;
padding: 10px 0;
        }
        nav ul {            list-style-
type: none;             margin: 0;
padding: 0;             text-align:
center;
        }         nav ul li {
display: inline;
margin: 0 20px;
        }         nav ul li a {             color:
white;             text-decoration: none;
font-size: 1.2em;             padding:
10px 20px;             transition:
background 0.3s;
        }
        nav ul li a:hover {             background-
color: #007bff;             border-radius: 5px;
        }         img{             max-
width: min-content;
        }
        .buttons {             text-
align:                    center;
margin: 30px 0;
        }
        .btn {
            background: #007bff;
            color: #fff;
border: none;
padding: 15px 30px;
margin: 10px;
border-radius: 8px;
```

```
font-size: 1.2em;
cursor: pointer;
            transition: background 0.3s, transform 0.3s;
        }
        .btn:hover                    {
background:              #0056b3;
transform: scale(1.05);
        }
        .info-section {
display: none;
margin-top: 30px;
        }
        .info-section h2 {
font-size: 2em;            color:
#007bff;            margin-
bottom: 20px;            text-
align: center;
        }
        .info-section p {
font-size: 1.1em;            line-
height: 1.8;            color:
#495057;            text-align:
justify;        }
        .info-section ul {
margin: 15px 0;
padding-left: 20px;
color: #495057;
        }
        .info-section ul li {
font-size: 1em;            line-height:
1.6;
        }
```

```
.info-section img {
width: 100%; height:
auto; border-radius:
8px; margin: 20px 0;
}
.register-btn { text-
align: center; margin-
top: 20px;
}
.register-btn button {
background: #007bff; color:
white; border: none;
padding: 12px 30px; font-
size: 1.1em; font-weight:
bold; text-transform:
uppercase; border-radius:
8px;
cursor: pointer;
transition: background 0.3s, transform 0.3s;
}
.register-btn button:hover {
background: #007bff; transform:
scale(1.05);
}
.info-section .apply-btn {
text-align: center; margin-top:
20px;
}
.info-section .apply-btn button {
padding: 12px 25px; font-size: 1em;
color: white; background-color:
#0056b3; border: none;
```

```
border-radius: 5px;            cursor: pointer;
transition: background-color 0.3s;
        }
        .info-section .apply-btn button:hover {            background-
color: #0056b3;
        }
    </style>
</head>
<body>
    <nav>
        <ul>
            <li><a href="#">Home</a></li>
            <li><a href="#hostel-info">Hostel Info</a></li>
            <li><a href="#mess-info">Mess Info</a></li>
            <li><a href="#">Contact</a></li>
        </ul>
    </nav>

    <div class="container">
        <header>
            <h1>Hostel and Mess Portal</h1>
            <p>Welcome to the Hostel and Mess Portal. Here you can access
important information regarding hostel allocations and mess services. Use the
buttons below to navigate to the respective sections.</p>
        </header>

        <section class="buttons">
            <button id="hostel-btn" class="btn">Hostel Information</button>
            <button id="mess-btn" class="btn">Mess Information</button>
</section>

        <!-- Hostel Information Section -->
        <section id="hostel-info" class="info-section">
            <h2>Hostel Admission Information</h2>
```

```
<p>Our hostel provides a safe, comfortable, and student-friendly environment, ensuring a balance between academics and personal growth. Below are the key features and details:</p>
<ul>
  <li>Total Rooms: 100</li>
  <li>Fees Per Student: Rs. 20,000 per semester</li>
  <li>Available Rooms: 30</li>
  <li>Features:
    <ul>
      <li>24/7 Security and Surveillance</li>
      <li>Fully Furnished Rooms with Wi-Fi</li>
      <li>On-site Laundry Facilities</li>
      <li>Recreational Areas and Sports Facilities</li>
      <li>Hygienic and Nutritious Food</li>
    </ul>
  </li>
</ul>
<img src="WhatsApp Image 2024-12-16 at 13.50.17_d061f50c.jpg" alt="Sample Hostel Photo">
<img src="C:\HMP\WhatsApp Image 2024-12-16 at 13.50.17_c443f98c.jpg" alt="Sample Hostel Photo">
<img src="C:\HMP\WhatsApp Image 2024-12-16 at 13.50.16_9acefbc7.jpg" alt="Sample Hostel Photo">
<img src="C:\HMP\WhatsApp Image 2024-12-16 at 13.50.16_3a4e94dd.jpg" alt="Sample Hostel Photo">
<div class="register-btn">
  <button onclick="window.location.href='host.html';">Register Now</button>
</div>
</section>
```

LIMITATIONS OF THE SYSTEM

While the Integrated Hostel and Mess Solution provides numerous benefits in streamlining and automating hostel and mess operations, there are certain limitations associated with the system. These limitations need to be considered for a more comprehensive evaluation of the project.

Here are the main limitations of the system:

Dependence on Internet Connectivity:

Since the system is web-based, it requires a stable internet connection to access and use the platform. Any network issues or outages can prevent users from accessing critical features like room booking, meal ordering, or payment processing.

Initial Setup Costs:

Although the system is cost-effective in the long run, the initial setup may involve some costs related to server hosting, software development, training staff, and integrating the system into existing workflows. These initial costs may be prohibitive for smaller institutions or private hostels.

Limited Customization for Specific Needs:

The current version of the system may not fully cater to the unique needs of different hostels or messes, especially if they have highly specific operational requirements. Customizing the system to address specific needs might require additional development, which can increase costs and time.

User Adoption and Training:

For staff members and students who are not tech-savvy, there might be a learning curve when it comes to using the platform. Proper training and orientation would be required to ensure smooth adoption. Resistance to change from users accustomed to manual processes might hinder full utilization of the system.

PROPOSED ENHANCEMENTS

Implement Mobile Application Support

Objective: Expand the system's accessibility by developing mobile applications for both Android and iOS platforms.

Rationale: Many users prefer accessing web applications via their mobile devices. Providing a dedicated mobile app will enhance user experience and increase accessibility for students and hostel managers.

Improve Offline Functionality

Objective: Enable key features of the system to work offline, allowing users to continue their tasks even when there's no internet connection.

Rationale: Many institutions in remote areas may experience unstable internet connections. Having offline functionality ensures that operations are not disrupted, and data is synced when the connection is restored.

Multilingual Support

Objective: Introduce multilingual capabilities to cater to users from different linguistic backgrounds.

Rationale: Institutions in diverse regions often have students and staff who speak different languages. Supporting multiple languages will make the system more inclusive.

CONCLUSION

The Integrated Hostel and Mess Solution project is designed to streamline and automate the operations of hostels and messes, providing a centralized system for managing various aspects such as room allocation, meal bookings, attendance tracking, payments, and reporting. With an emphasis on ease of use and efficiency, the system addresses the challenges faced by students, hostel administrators, and mess owners in managing daily tasks and resources.

The system effectively integrates several critical modules, including room management, meal planning, payment systems, and attendance tracking. By automating these processes, the solution reduces administrative overhead and ensures accurate record-keeping. This not only improves operational efficiency but also enhances the overall user experience for students, making it easier to track and manage their hostel and mess-related activities.

The proposed enhancements for the system, such as mobile app support, offline functionality, multilingual support, and advanced data security measures, aim to further improve the system's usability and accessibility. Integrating with third-party systems, like biometric attendance devices and payment gateways, will provide a seamless experience for both users and administrators.

BIBLIOGRAPHY

Websites: Python Official Documentation: https://www.python.org/doc/

- Django Framework Documentation: https://docs.djangoproject.com/en/stable/
- W3Schools: https://www.w3schools.com/

Miscellaneous References

- Django Project: https://www.djangoproject.com/
- SQLite Documentation: https://www.sqlite.org/docs.html

Online Resources

- GeeksforGeeks: https://www.geeksforgeeks.org/django-tutorial/
- Stack Overflow: https://stackoverflow.com/